

Module 10 – Big Ideas: Dashboards, Machine Learning & Capstone Delivery

Big Idea 1: Data Products > Notebooks

Up until now, most of your work has lived in notebooks.

In this module, you move toward **data products**:

- Interactive dashboards
- Maps
- Models that make predictions

Key shift:

From:

“Here is my analysis”

To:

“Here is something you can use”

Examples:

- A dashboard where users filter launch data
- A map showing launch site patterns
- A model predicting landing success

Why it matters

Stakeholders don't read notebooks

They interact with tools and outputs

Big Idea 2: Interactivity Makes Data Useful

Static charts are helpful, but limited

Interactive analytics lets users:

- Filter data (launch site, payload range)
- Explore patterns on their own
- Ask “what if” questions

In this module:

- Folium → geographic exploration
- Plotly Dash → interactive dashboards

Key idea:

Good dashboards answer questions repeatedly

Why it matters

You are building something reusable, not one-off

Big Idea 3: Dashboards = Logic + UI

Dashboards are not just charts

They combine:

- Frontend (layout, controls, visuals)
- Backend logic (callbacks, filtering, transformations)

Core concepts:

- Layout → what the user sees
- Inputs → dropdowns, sliders
- Callbacks → how the app reacts to changes

Key idea:

A dashboard is an application, not a plot

Why it matters

This is how modern analytics tools are built

Big Idea 4: Machine Learning = Prediction Under Uncertainty

Now you actually build a predictive model

Goal:

Predict whether a SpaceX rocket will successfully land

Core workflow:

- Feature preparation
- Train/test split
- Model training
- Hyperparameter tuning

- Evaluation

Models explored:

- Logistic Regression
- SVM
- KNN
- Tree-based models

Key idea:

Models are not truth, they are approximations

Why it matters

You are turning patterns into decisions

Big Idea 5: Evaluation > Accuracy

Not all models are equal

You evaluate using:

- Confusion matrix
- Precision / Recall
- F1 score
- ROC (optional)

Key idea:

Accuracy alone can be misleading

Example:

- Predicting all launches succeed could look “accurate”
- But completely useless

Why it matters

You need to measure what actually matters for the problem

Big Idea 6: Model Tuning and Fair Comparison

You don’t just train one model

You:

- Try multiple algorithms
- Tune hyperparameters (GridSearchCV)
- Compare results fairly

Key idea:

Model performance depends heavily on configuration

Also:

- Never peek at the test set during tuning
- Keep evaluation honest

Why it matters

Bad evaluation leads to false confidence

Big Idea 7: Persistence Connects to the Real World

Training a model is not the end

You can:

- Save the model (e.g., pickle)
- Reload it later
- Use it for predictions

Key idea:

Models need to be reusable

Why it matters

This is how models move from notebooks → production

Big Idea 8: Communication Is the Final Step

The project is not complete until it is communicated

Deliverables:

- GitHub repo (code + outputs)
- Slide deck
- Written report

Key idea:

Data science = communication

Your job:

- Explain what you did
- Explain what you found
- Explain why it matters

Why it matters

Insights only matter if people understand them

Big Idea 9: Storytelling Turns Analysis Into Decisions

You are not just presenting results

You are telling a story:

- Problem → Data → Analysis → Model → Results → Impact

Examples:

- Why launch site matters
- How success rates changed over time
- What drives landing success

Key idea:

Structure matters more than complexity

Why it matters

This is what stakeholders remember

Big Idea 10: End-to-End Data Science

This is the full pipeline:

Data Collection → Wrangling → EDA → Modeling → Visualization → Communication

Everything connects

Key idea:

Data science is a system, not steps

Why it matters

This is the closest thing to real-world work in the course